

Interview with Lynn Anderson, an employee of ConAm Inspection
Interview conducted on September 29, 2006
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Summary of NTSB Interview of: Lynn Anderson
Date of Interview: September 29, 2006
Time of Interview: 9:30 a.m.
Location of Interview: 2 Millbury Street
Auburn, MA

On above date and time, investigators with the NTSB interviewed Mr. **LYNN ANDERSON** in regard to the post-installation testing of epoxy anchors in the Interstate 90 (I-90) Connector Tunnel. During the time of these tests, Mr. Anderson was employed by ConAm Inspection Inc., (a subsidiary of Stavely Inc.) as a certified welding inspector (CWI). As an employee with ConAm Inspection, he had performed load testing since 1988.

Mr. Anderson was unable to recall the dates or the overall length of time during which he performed the post-installation load tests of the epoxy anchors, but did state that the tests were done within the I-90 Connector Tunnel. Also, he was not able to recall or describe the exact locations within the tunnel where he conducted tests.

According to Mr. Anderson, during the time frame in which these tests were performed, there was no regularity in the scheduling of the tests. ConAm would reportedly receive a phone call to notify them of the need to test a group of anchors. As such, Modern Continental Construction (MCC) would always notify ConAm ahead of time as to when their services were needed. ConAm would then arrange for Mr. Anderson to be available to conduct the tests.

Mr. Anderson described a typical work day as starting at 7:00 a.m. and finishing at 3:30 p.m. Travel time to the job site was included as part of the work day hours and on some days he would work through his lunch hour. During the testing process, Mr. Anderson worked with another ConAm employee whose job was to assist him by mounting the testing fixture, applying pressure to the load cell, and repositioning the fixture to another anchor once a test was concluded. During the actual tests, Mr. Anderson's job was to monitor and record the results.

The specifications for the test fixture were provided for in the construction contract. Mr. Anderson described the fixture as having a triangular base, a manually operated (hand pump) hydraulic ram, and a pressure gauge to measure the amount of force being applied to the ram.

Mr. Anderson described the events during a typical day of testing as follows:

- He and his assistant would arrive at the work site and meet with an engineer from MCC. The engineer would give them their assignment regarding which anchors were to be tested.
- Using a motorized scissors lift, Mr. Anderson and his assistant would access the ceiling mounted anchors and the assistant would thread a rod onto one of the anchors. The test fixture was then placed over the rod and secured by means of a nut and washers.

- The assistant would then pump the test fixture's hydraulic ram to a specified load, which Mr. Anderson recalled as usually being in the vicinity of 3,200 pounds¹.
- During each test the load was to be applied for two minutes, during which time Mr. Anderson would visually observe both the fixture's gauge and the anchor for any signs of movement. If no movement was observed the anchor was considered to have passed the test and was marked to indicate that it was "OK". However, if movement was detected, the anchor was deemed to have failed and was subsequently removed from the ceiling.
- Mr. Anderson estimated that it took between three to four minutes to test each anchor. Also, engineers from Bechtel Parsons Brinckerhoff (B/PB) would periodically show up throughout the day to observe the tests as they were being performed. He also stated that the BPB employees would measure the duration of the tests.
- After each test was performed, Mr. Anderson would record the results. Later, the results were entered onto a form and submitted to MCC. Mr. Anderson believed that MCC tracked the number of tests that were conducted on a given day, as well as the number of anchors that passed or failed.

In addition to the typical anchor tests that Mr. Anderson had described, he also recalled performing some tests after the concrete ceiling panels had been set into place. As with the other tests, these were done at the direction of MCC. In order to complete these load tests, Mr. Anderson and his assistant were required to get on top of the already hanging ceiling panels. To gain access to the anchors selected for testing, personnel from MCC would remove the associated bracket attached by the anchors to the tunnel's concrete ceiling. These brackets were used to secure the tie rods attached to the back of the ceiling panels and ultimately secured the ceiling panels above the roadway. As such, Mr. Anderson reported that during the testing process, MCC personnel would only remove one bracket at a time. Mr. Anderson was not able to recall why these tests were performed or the number of these types of tests that were conducted.

¹ Due to the amount of time that had elapsed since the testing was done, Mr. Anderson was unable to recall the exact load that was placed on the anchors. He also stated that during this time, some of the anchors were tested at a higher load which he believed was about 6,300 pounds.